Ethiopian E-Commerce Platform

Advanced Database Systems — Group 2 Project

Entity-Relationship Diagram & Relational Schema

TOTALTABLES
35+

NORMALIZATION
3NF / BCNF

Target : MySQL 8.0+ (InnoDB, utf8mb4, ROW_FORMAT=DYNAMIC)

Scale : Designed for millions of users, multi-city Ethiopia operations

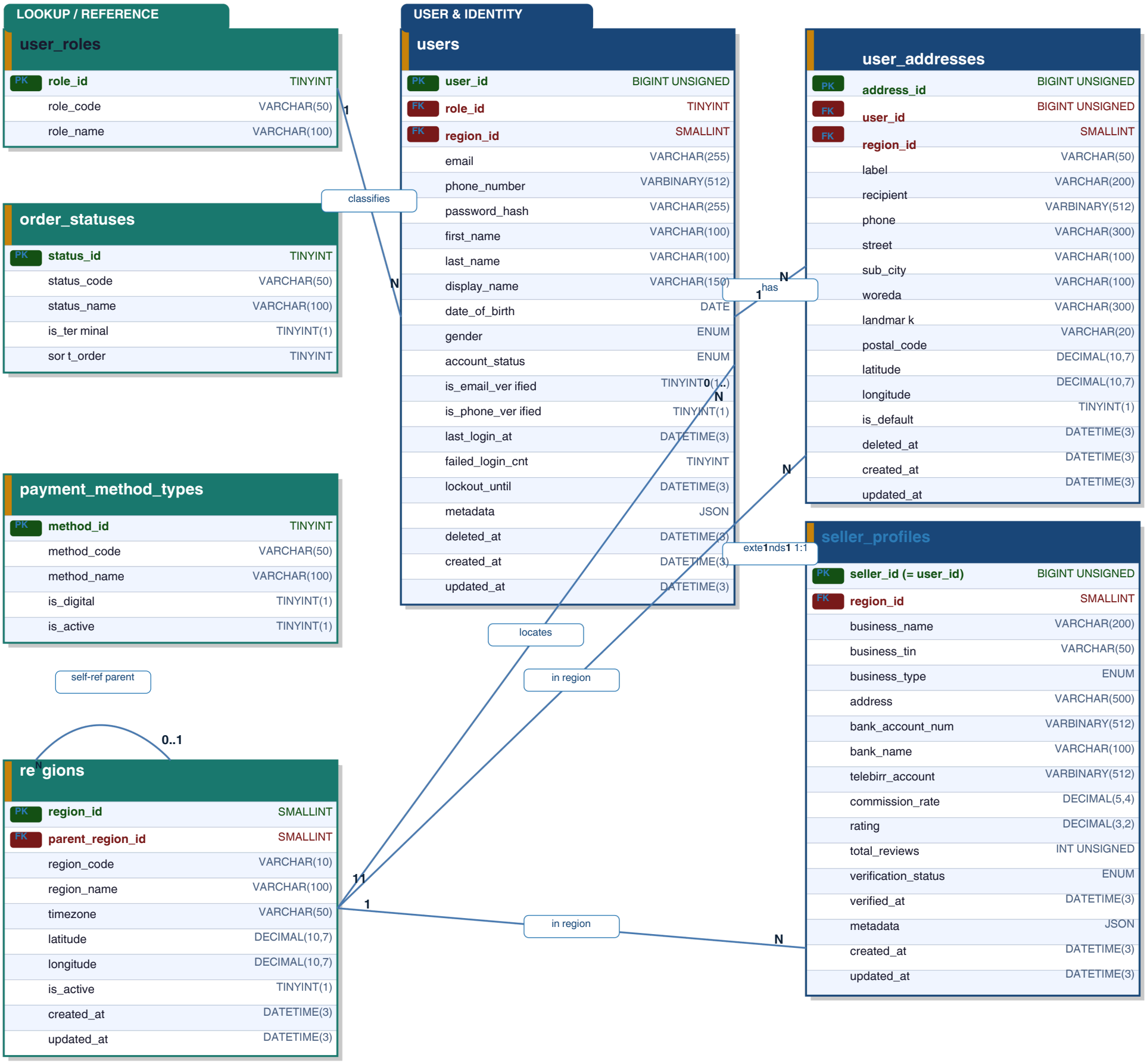
Cities :
Addis Ababa, Adama, Hawassa, Dire Dawa, Mekele, Bahir Dar .....

SUBSYSTEMS
9 Functional Domains

DIAGRAM PAGES
8 Pages

DATABASE SUBSYSTEMS

- Lookup/Reference
- User & Identity
- Product Catalog
- Inventory
- Orders
- Payments
- Security & Audit
- Analytics
- Social & Promo


LOOKUP / REFERENCE
user_roles
PK role_id TINYINT
role_code VARCHAR(50)
role_name VARCHAR(100)
order_statuses
PK status_id TINYINT
status_code VARCHAR(50)
status_name VARCHAR(100)
is_ter minal TINYINT(1)
sor t_order TINYINT
payment_method_types
PK method_id TINYINT
method_code VARCHAR(50)
method_name VARCHAR(100)
is_digital TINYINT(1)
is_active TINYINT(1)
self-ref parent
0..1
re gions
PK region_id SMALLINT
FK parent_region_id SMALLINT
region_code VARCHAR(10)
region_name VARCHAR(100)
timezone VARCHAR(50)
latitude DECIMAL(10,7)
longitude DECIMAL(10,7)
is_active TINYINT(1)
created_at DATETIME(3)
updated_at DATETIME(3)
USER & IDENTITY
users
PK user_id BIGINT UNSIGNED
FK role_id TINYINT
FK region_id SMALLINT
email VARCHAR(255)
phone_number VARBINARY(512)
password_hash VARCHAR(255)
first_name VARCHAR(100)
last_name VARCHAR(100)
display_name VARCHAR(150)
date_of_birth DATE
gender ENUM
account_status ENUM
is_email_ver ified TINYINT(1)
is_phone_ver ified TINYINT(1)
last_login_at DATETIME(3)
failed_login_cnt TINYINT
lockout_until DATETIME(3)
metadata JSON
deleted_at DATETIME(3)
created_at DATETIME(3)
updated_at DATETIME(3)
user_addresses
PK address_id BIGINT UNSIGNED
FK user_id BIGINT UNSIGNED
FK region_id SMALLINT
label VARCHAR(50)
recipient VARCHAR(200)
phone VARBINARY(512)
street VARCHAR(300)
sub_city VARCHAR(100)
woreda VARCHAR(100)
landmar k VARCHAR(300)
postal_code VARCHAR(20)
latitude DECIMAL(10,7)
longitude DECIMAL(10,7)
is_default TINYINT(1)
deleted_at DATETIME(3)
created_at DATETIME(3)
updated_at DATETIME(3)
seller_profiles
PK seller_id (= user_id) BIGINT UNSIGNED
FK region_id SMALLINT
business_name VARCHAR(200)
business_tin VARCHAR(50)
business_type ENUM
address VARCHAR(500)
bank_account_num VARBINARY(512)
bank_name VARCHAR(100)
telebirr_account VARBINARY(512)
commission_rate DECIMAL(5,4)
rating DECIMAL(3,2)
total_reviews INT UNSIGNED
verification_status ENUM
verified_at DATETIME(3)
metadata JSON
created_at DATETIME(3)
updated_at DATETIME(3)
classifies
1
N
has
N
1
exte1nds1 1:1
N
locates
N
in region
1
1
in region
N

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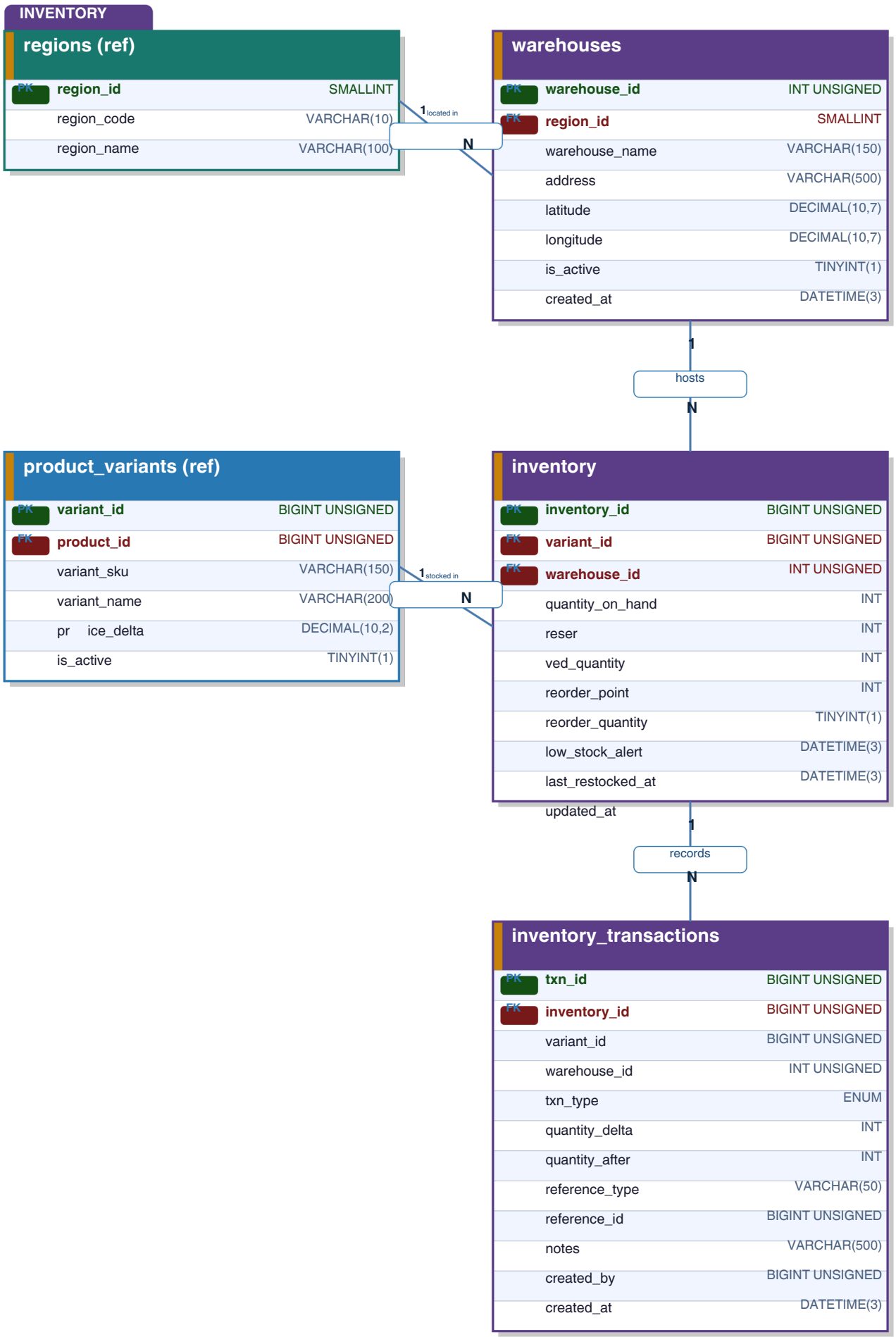
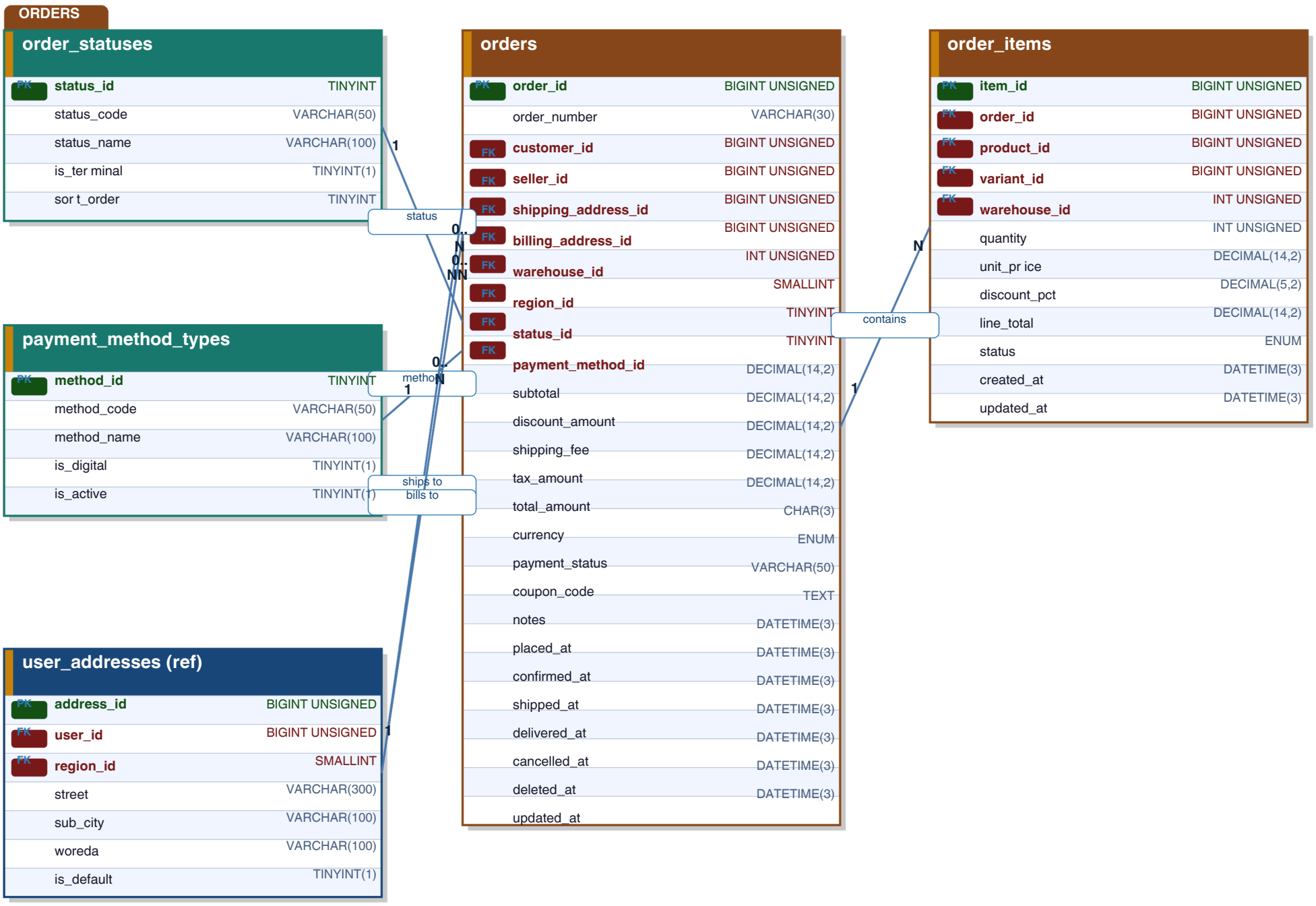
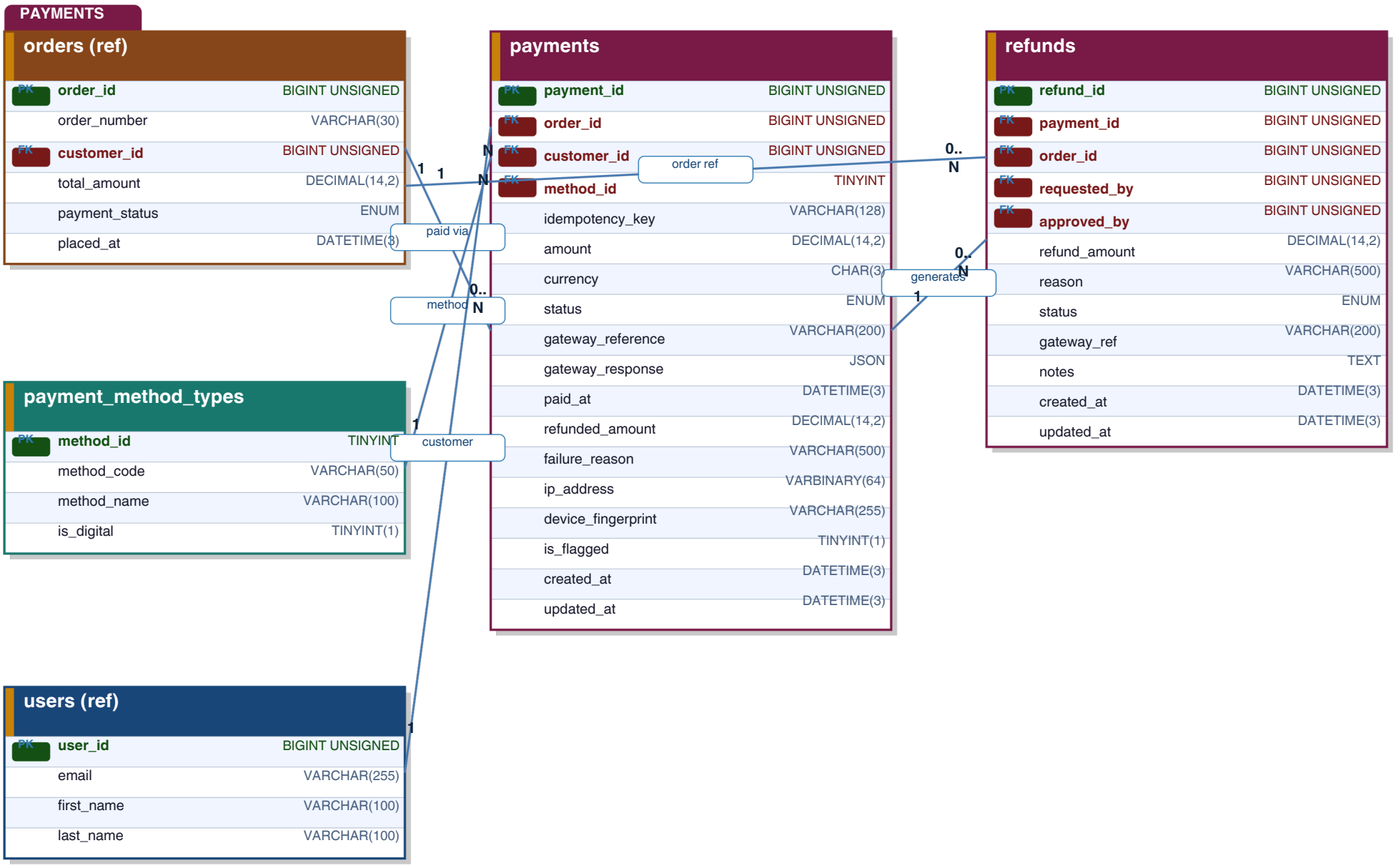


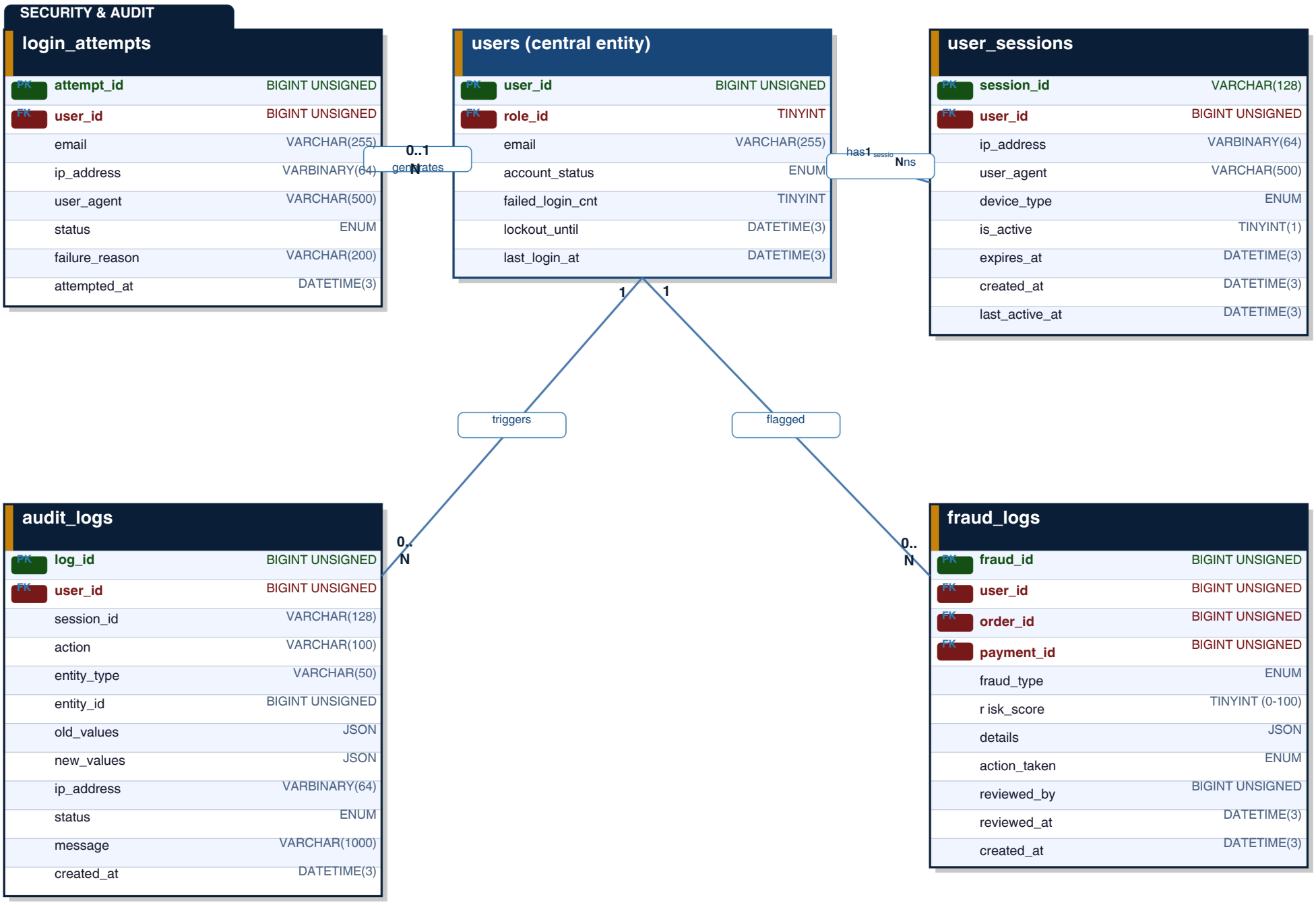
UNIQUE Constraint
(variant_id, warehouse_id)
One stock record per variant
per warehouse location

CHECK Constraint
quantity_on_hand "e reserved_quantity
(available stock must be "e 0)


ORDERS
order_statuses
status_id TINYINT
status_code VARCHAR(50)
status_name VARCHAR(100)
is_ter minal TINYINT(1)
sor t_order TINYINT
orders
order_id BIGINT UNSIGNED
order_number VARCHAR(30)
customer_id BIGINT UNSIGNED
seller_id BIGINT UNSIGNED
shipping_address_id BIGINT UNSIGNED
billing_address_id BIGINT UNSIGNED
warehouse_id INT UNSIGNED
region_id SMALLINT
status_id TINYINT
payment_method_id TINYINT
subtotal DECIMAL(14,2)
discount_amount DECIMAL(14,2)
shipping_fee DECIMAL(14,2)
tax_amount DECIMAL(14,2)
total_amount DECIMAL(14,2)
currency CHAR(3)
payment_status ENUM
coupon_code VARCHAR(50)
notes TEXT
placed_at DATETIME(3)
confirmed_at DATETIME(3)
shipped_at DATETIME(3)
delivered_at DATETIME(3)
cancelled_at DATETIME(3)
deleted_at DATETIME(3)
updated_at DATETIME(3)
order_items
item_id BIGINT UNSIGNED
order_id BIGINT UNSIGNED
product_id BIGINT UNSIGNED
variant_id BIGINT UNSIGNED
warehouse_id INT UNSIGNED
quantity INT UNSIGNED
unit_pr ice DECIMAL(14,2)
discount_pct DECIMAL(5,2)
line_total DECIMAL(14,2)
status ENUM
created_at DATETIME(3)
updated_at DATETIME(3)
payment_method_types
method_id TINYINT
method_code VARCHAR(50)
method_name VARCHAR(100)
is_digital TINYINT(1)
is_active TINYINT(1)
user_addresses (ref)
address_id BIGINT UNSIGNED
user_id BIGINT UNSIGNED
region_id SMALLINT
street VARCHAR(300)
sub_city VARCHAR(100)
woreda VARCHAR(100)
is_default TINYINT(1)
status
method
ships to
bills to
contains
1
0..N
N
1


PAYMENTS
orders (ref)
PK order_id BIGINT UNSIGNED
order_number VARCHAR(30)
FK customer_id BIGINT UNSIGNED
total_amount DECIMAL(14,2)
payment_status ENUM
placed_at DATETIME(3)
payments
PK payment_id BIGINT UNSIGNED
FK order_id BIGINT UNSIGNED
FK customer_id BIGINT UNSIGNED
FK method_id TINYINT
idempotency_key VARCHAR(128)
amount DECIMAL(14,2)
currency CHAR(3)
status ENUM
gateway_reference VARCHAR(200)
gateway_response JSON
paid_at DATETIME(3)
refunded_amount DECIMAL(14,2)
failure_reason VARCHAR(500)
ip_address VARBINARY(64)
device_fingerprint VARCHAR(255)
is_flagged TINYINT(1)
created_at DATETIME(3)
updated_at DATETIME(3)
refunds
PK refund_id BIGINT UNSIGNED
FK payment_id BIGINT UNSIGNED
FK order_id BIGINT UNSIGNED
FK requested_by BIGINT UNSIGNED
FK approved_by BIGINT UNSIGNED
refund_amount DECIMAL(14,2)
reason VARCHAR(500)
status ENUM
gateway_ref VARCHAR(200)
notes TEXT
created_at DATETIME(3)
updated_at DATETIME(3)
payment_method_types
PK method_id TINYINT
method_code VARCHAR(50)
method_name VARCHAR(100)
is_digital TINYINT(1)
users (ref)
PK user_id BIGINT UNSIGNED
email VARCHAR(255)
first_name VARCHAR(100)
last_name VARCHAR(100)
order ref
paid via
method
customer
generates
1
N
0..
N

SECURITY & AUDIT



& Append-Only Tables

login_attempts and audit_logs are immutable — never UPDATE or DELETE. Partition by month for performance.

ANALYTICS

daily_sales_summary

PK	summar y_date	DATE
PK	region_id	SMALLINT
PK	seller_id	BIGINT UNSIGNED
	total_orders	INT UNSIGNED
	total_revenue	DECIMAL(18,2)
	total_items_sold	INT UNSIGNED
	total_refunds	DECIMAL(18,2)
	avg_order_value	DECIMAL(14,2)
	updated_at	DATETIME(3)

product_sales_stats

PK	product_id	BIGINT UNSIGNED
PK	period_date	DATE
	units_sold	INT UNSIGNED
	revenue	DECIMAL(14,2)
	retur ns	INT UNSIGNED
	views	INT UNSIGNED
	updated_at	DATETIME(3)

customer_ltv

PK	customer_id	BIGINT UNSIGNED
	first_order_date	DATE
	last_order_date	DATE
	total_orders	INT UNSIGNED
	total_spent	DECIMAL(18,2)
	total_refunds	DECIMAL(18,2)
	avg_order_value	DECIMAL(14,2)
	ltv_score segment	DECIMAL(10,2)
	updated_at	ENUM
		DATETIME(3)

SOCIAL & PROMO

product_reviews

PK	review_id	BIGINT UNSIGNED
FK	product_id	BIGINT UNSIGNED
FK	customer_id	BIGINT UNSIGNED
FK	order_id	BIGINT UNSIGNED
	rating	TINYINT (1-5)
	title	VARCHAR(200)
	body	TEXT
	is_ver ified	TINYINT(1)
	is_approved	TINYINT(1)
	helpful_votes	INT UNSIGNED
	deleted_at	DATETIME(3)
	created_at	DATETIME(3)

coupons

PK	coupon_id	INT UNSIGNED
	coupon_code	VARCHAR(50)
FK	created_by	BIGINT UNSIGNED
	discount_type	ENUM
	discount_value	DECIMAL(10,2)
	min_order_amt	DECIMAL(10,2)
	max_discount	DECIMAL(10,2)
	usage_limit	INT UNSIGNED
	usage_count	INT UNSIGNED
	per_user_limit	INT UNSIGNED
	valid_from	DATETIME(3)
	valid_until	DATETIME(3)
	is_active	TINYINT(1)
	created_at	DATETIME(3)

notifications

PK	notification_id	BIGINT UNSIGNED
FK	user_id	BIGINT UNSIGNED
	type	VARCHAR(100)
	channel	ENUM
	title	VARCHAR(200)
	body	TEXT
	data	JSON
	is_read	TINYINT(1)
	sent_at	DATETIME(3)
	read_at	DATETIME(3)
	created_at	DATETIME(3)

DISTRIBUTED

shard_map

PK	shard_id	TINYINT UNSIGNED
	shard_name	VARCHAR(50)
	dsn_pr imar y	VARCHAR(200)
	dsn_replica	VARCHAR(200)
	region_ids	JSON
	is_active	TINYINT(1)
	max_user_id	BIGINT UNSIGNED
	min_user_id	BIGINT UNSIGNED

Distributed Architecture

- Horizontal fragmentation by region_id
- Shard 1 !' Addis Ababa (primary)
- Shard 2 !' Adama
- Shard 3 !' Hawassa
- dsn_primary / dsn_replica: connection strings for each regional shard node
- region_ids JSON maps regions to shards